

RATIOS, RATES & PROPORTIONS

Digital SAT Style Practice Questions

DOCUMENT 1 — QUESTIONS ONLY

Easy 25% (10) | Medium 45% (18) | Hard 30% (12) | Total Questions: 40

Q1. • EASY

A recipe uses 8 cups of flour for every 12 cups of sugar. What is the ratio of flour to sugar, written in simplest form?

- A) 4 : 6
- B) 2 : 3
- C) 3 : 2
- D) 8 : 12

Q2. • EASY

If $\frac{3}{4} = \frac{x}{20}$, what is the value of x ?

- A) 16
- B) 12
- C) 15
- D) 26.67

Q3. • EASY

A car travels 240 miles in 4 hours at a constant speed. What is the car's speed, in miles per hour?

- A) $\frac{1}{60}$
- B) 4
- C) 240
- D) 60

Q4. • EASY

A store sells 3 pounds of coffee beans for \$15. What is the price per pound?

- A) \$5
- B) \$0.20
- C) \$18
- D) \$45

Q5. • EASY GRID-IN

If $5/8 = x/56$, what is the value of x ?

GRID-IN — enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.

Q6. • EASY

On a scale drawing, 1 inch represents 5 feet of actual distance. If two points are 7 inches apart on the drawing, how many feet apart are they in reality?

- A) 12
 - B) 35
 - C) 1.4
 - D) 0.71
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Q7. • EASY

A jar contains red and blue marbles in the ratio 3 : 5. If the jar contains 40 marbles total, how many are red?

- A) 24
 - B) 25
 - C) 15
 - D) 8
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Q8. • EASY

A machine produces 150 widgets in 6 hours, working at a constant rate. How many widgets does it produce per hour?

- A) 900
 - B) 144
 - C) 6
 - D) 25
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Q9. • EASY GRID-IN

A recipe that serves 4 people uses 6 eggs. At the same ratio, how many eggs are needed to serve 10 people?

GRID-IN — enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.

Q10. • EASY

In a class, the ratio of boys to girls is 5 : 7. If there are 36 students total, how many are girls?

- A) 21
- B) 15

- C) 36
 - D) 25.2
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Q11. • MEDIUM

In a survey, the ratio of A to B is 2 : 3, and the ratio of B to C is 4 : 5. What is the ratio of A to C?

- A) 6 : 20
 - B) 8 : 15
 - C) 2 : 5
 - D) 3 : 10
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Q12. • MEDIUM

On a map, the scale is 1 : 50,000 (1 unit on the map represents 50,000 of the same unit in reality). If two cities are 3.4 centimeters apart on the map, how far apart are they in reality, in kilometers?

- A) 170
 - B) 17
 - C) 1.7
 - D) 0.17
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Q13. • MEDIUM GRID-IN

Pipe A can fill a tank in 6 hours working alone. Pipe B can fill the same tank in 12 hours working alone. If both pipes work together, how many hours will it take to fill the tank?

GRID-IN — enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.

Q14. • MEDIUM

Solution A is 20% salt by volume, and Solution B is 50% salt by volume. If 4 liters of Solution A are mixed with 2 liters of Solution B, what percent of the resulting mixture is salt?

- A) 35%
 - B) 25%
 - C) 70%
 - D) 30%
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Q15. • MEDIUM

A car travels at a constant speed of 45 miles per hour. What is this speed in feet per second? (1 mile = 5,280 feet)

- A) 66

- B) 45
- C) 33
- D) 108

Q16. • MEDIUM

y varies directly with x. When $x = 4$, $y = 20$. What is the value of y when $x = 9$?

- A) 36
- B) 45
- C) 20
- D) 11.25

Q17. • MEDIUM GRID-IN

A recipe for 6 servings requires 2.5 cups of rice. At the same ratio, how many cups of rice are needed for 15 servings?

GRID-IN — enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.

Q18. • MEDIUM

A town has a population of 18,000 people and covers an area of 12 square miles. What is the town's population density, in people per square mile?

- A) 0.00067
- B) 216,000
- C) 1,500
- D) 150

Q19. • MEDIUM

Runner A completes 100 meters in 12 seconds. Runner B completes 150 meters in 17 seconds. Based on average speed, which runner is faster?

- A) Runner A, because 12 is less than 17
- B) The two runners are equally fast
- C) Runner A, because 100 is less than 150
- D) Runner B

Q20. • MEDIUM

At a company, the ratio of managers to employees is 1 : 12. If the company has 156 employees, how many managers does it have?

- A) 13

- B) 12
- C) 156
- D) 1,872

Q21. • MEDIUM

A printing company charges customers based on a constant rate per page. Printing 250 pages costs \$32.50. At this same rate, how much would it cost to print 880 pages?

- A) \$104.00
- B) \$114.40
- C) \$32.50
- D) \$6,765.63

Q22. • MEDIUM GRID-IN

If 3 workers can assemble 18 chairs in a day, how many chairs can 7 workers assemble in a day, working at the same individual rate?

GRID-IN — enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.

Q23. • MEDIUM

A water tank fills at a constant rate of 2.5 gallons per minute. How many gallons does it fill in 1 hour and 12 minutes?

- A) 150
- B) 175
- C) 180
- D) 30

Q24. • MEDIUM

In a classroom, the ratio of boys to girls is 3 : 4. If there are 15 more girls than boys, how many boys are in the classroom?

- A) 15
- B) 20
- C) 60
- D) 45

Q25. • MEDIUM

Pipe A alone fills a pool in 8 hours. Working together, pipe A and pipe B fill the same pool in 3 hours. How long would it take pipe B alone to fill the pool?

- A) 4.8
 - B) 5
 - C) 11
 - D) 2.67
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Q26. • MEDIUM ✎ GRID-IN

A train travels 315 miles in 4.5 hours at a constant speed. At this same speed, how many miles would the train travel in 7 hours?

GRID-IN — enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.

Q27. • MEDIUM

Brand A sells a 16-ounce bag of pasta for \$3.20. Brand B sells a 24-ounce bag of the same type of pasta for \$4.32. Which brand offers the lower price per ounce, and what is that price?

- A) Brand A, \$0.20 per ounce
 - B) Brand B, \$0.18 per ounce
 - C) Brand A, \$0.18 per ounce
 - D) Brand B, \$0.20 per ounce
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Q28. • MEDIUM

A 750-milliliter bottle of juice concentrate contains 250 milliliters of pure juice; the rest of the bottle is water. What is the ratio of pure juice to water in the bottle, written in simplest form?

- A) 1 : 3
 - B) 1 : 1
 - C) 1 : 2
 - D) 2 : 1
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Q29. • HARD

In a fruit basket, the ratio of apples to oranges is 3 : 4, and the ratio of oranges to pears is 4 : 5. If the basket contains 180 pieces of fruit total, how many pears are there?

- A) 60
 - B) 48
 - C) 45
 - D) 75
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Q30. • HARD

Working alone, Maria can paint a fence in 8 hours. Working alone, Jamal can paint the same fence in 12 hours. They work together for 3 hours, after which Maria leaves and Jamal finishes the job alone. How many more hours does Jamal need to finish?

- A) 4.5
- B) 5
- C) 9
- D) 2.4

Q31. • **HARD** **GRID-IN**

How many liters of an 80% acid solution must be mixed with 5 liters of a 20% acid solution to create a 50% acid solution?

GRID-IN — enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.

Q32. • **HARD**

A cyclist rides from town A to town B at a constant speed of 12 miles per hour, then immediately returns from town B to town A along the same route at a constant speed of 18 miles per hour. What is the cyclist's average speed, in miles per hour, for the entire round trip?

- A) 15
- B) 14.4
- C) 13
- D) 16

Q33. • **HARD**

A factory uses Machine X, which produces 45 parts per hour, and Machine Y, which produces 60 parts per hour. Both machines run together for 4 hours, after which Machine X breaks down and only Machine Y continues running for 2 more hours. How many total parts are produced?

- A) 420
- B) 480
- C) 540
- D) 630

Q34. • **HARD**

The ratio of x to y is 5 : 7. If x is increased by 10 and y is increased by 10, the new ratio of x to y becomes 3 : 4. What is the value of x ?

- A) 40
- B) 10
- C) 30
- D) 50

Q35. • **HARD** **GRID-IN**

It takes 5 workers 9 days to build a fence, with each worker working at the same constant individual rate. Working at that same individual rate, how many days would it take 15 workers to build the identical fence?

GRID-IN — enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.

Q36. • **HARD**

y is inversely proportional to x. When $x = 15$, $y = 8$. What is the value of y when $x = 24$?

- A) 5
 - B) 12.8
 - C) 9.375
 - D) 40
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Q37. • **HARD**

A scale model of a car is built at a scale of 1 : 18, meaning the model's length is $\frac{1}{18}$ of the actual car's length. If the actual car is 16.2 feet long, how many inches long is the model? (1 foot = 12 inches)

- A) 0.9
 - B) 10.8
 - C) 1.08
 - D) 324
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Q38. • **HARD**

Pipe A can fill an empty pool in 10 hours. Pipe B can fill the same pool in 15 hours. Pipe C, a drain, can empty a full pool in 12 hours. If all three pipes are open at the same time on an empty pool, how many hours will it take to fill the pool?

- A) 4
 - B) 7.5
 - C) 12
 - D) 27
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Q39. • **HARD** **GRID-IN**

At a school, the ratio of administrators to teachers is 1 : 6, and the ratio of teachers to students is 1 : 18. If there are 45 administrators, how many students are there?

GRID-IN — enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.

Q40. • **HARD**

A recipe that serves 24 people requires flour and sugar in the ratio 5 : 2, using a combined total of 4.2 cups of flour and sugar. If the recipe is scaled to serve 60 people, how many cups of sugar are needed?

- A) 1.2
- B) 7.5
- C) 1.5
- D) 3

— END OF QUESTIONS —